BChemXtract vs Open Babel

Structure-extraction performance over the BChemXtract `.cdx` test corpus, including BChemXtract v1.0 → v1.1.3

Generated 2026-07-08 · 200 ChemDraw binary files under `src/test/resources` · BChemXtract v1.0 & v1.1.3 vs Open Babel 3.1.0 · identical corpus & extraction path for all three

At a glance

CHEMICAL QUALITY	UNEXPANDED ALIASES	V1.0 → V1.1.3
100% vs 15%	**0 vs 533**	**hang fixed**
REAL INCHIS: BCHEMXTRACT VS OPEN BABEL	LEFTOVER `*` ATOMS: BCHEMXTRACT VS OPEN BABEL	1 FILE HANGS V1.0 · V1.1.3: ~50 MS

All three configurations extracted every structure from the same 200 binary ChemDraw (`.cdx`) files, deduplicating per file on InChIKey. Both BChemXtract versions produce chemically resolved output; Open Babel is far slower and mostly emits wildcard-bearing SMILES.

Comparison

Metric	BChemXtract v1.0	BChemXtract v1.1.3	Open Babel 3.1.0
COVERAGE			
Files in corpus	200	200	200
Files completed	**198**	200	200
Files yielding structures	181	179	199
Empty (0 found)	17	20	0
Failures (error / hang)	**2**	1	1
YIELD & CHEMICAL QUALITY			
Unique substances	1042 *	1029	1128
Real InChIs generated	**1042**	**1028**	**172**
SMILES fallback (no InChI)	0	1	956
Structures with unexpanded alias (`*`)	**0**	**0**	**533**
Fully expanded structures	1042	1029	595
SPEED			
Total time	4.6 s †	6.9 s	80.1 s
Average per file	23 ms	34 ms	401 ms
Throughput	43 files/s	29 files/s	2.5 files/s

> * v1.0 counts are over the 199 files it can process (excludes the file it hangs on). † v1.0 time is a warm single-JVM run over those 199 files; on the full corpus v1.0 **does not terminate** (see below). v1.1.3 and Open Babel times cover all 200 files.

> **Regression fixed between v1.0 and v1.1.3.** BChemXtract v1.0 hangs indefinitely on `volumeTest/bug/m17121576-9.cdx` — a whole-corpus run pinned one CPU core at 100% for over 30 minutes without progressing, and a per-file run was killed at a 60 s timeout. v1.1.3 extracts that same file in roughly 50 ms. This is the decisive difference between the two BChemXtract versions: chemical quality and per-file speed are otherwise equivalent.

Key findings

- **BChemXtract vs Open Babel — chemistry.** Both BChemXtract versions produce a genuine InChI for essentially every substance (v1.0: 1042/1042; v1.1.3: 1028/1029) and leave *no* unexpanded aliases. Open Babel produces real InChIs for only 172 of 1128 structures and leaves 533 carrying wildcard `*` atoms, because it cannot expand ChemDraw abbreviations/Sgroups — so most of its output is wildcard-bearing SMILES rather than identifiable substances.
- **BChemXtract vs Open Babel — speed.** BChemXtract processes the corpus in ~5–7 s; Open Babel takes 80 s (~12–17× slower), dominated by slow InChI generation.
- **v1.0 → v1.1.3 — robustness.** The one substantive change visible here is that v1.1.3 no longer hangs on `m17121576-9.cdx`. On every other file the two versions are effectively identical in output and per-file speed.
- **Shared hard failure.** All three fail on `volumeTest/vol20Test/m31717432-graphical-abstract.cdx` — it is not a valid binary CDX (no ChemDraw header).
- **Coverage difference is semantic.** The ~17–20 files BChemXtract reports as empty are S-group / copolymer / Markush fixtures it deliberately does not turn into discrete substances; Open Babel emits raw structures for them, inflating its file-coverage count.

Conclusion

On this 200-file ChemDraw corpus, both BChemXtract versions decisively outperform Open Babel on extraction quality — near-100 % real InChIs and complete alias expansion versus Open Babel's ~15 % InChIs and 533 unresolved wildcards — while running roughly an order of magnitude faster. Between the BChemXtract versions, chemical output and per-file speed are equivalent; the meaningful gain in v1.1.3 is robustness: it processes a file that hangs v1.0 indefinitely. For identifiable, InChI-grade substance extraction from ChemDraw files, BChemXtract v1.1.3 is the clear choice.

> **Methodology.** BChemXtract path (both versions): `CDXReader.readDocument` → `SubstanceXtractor.xtractUnique(doc, info, false)`; "real InChI" = non-empty `BCXSubstance.getInchi()`; "unexpanded alias" = SMILES containing `*`. The v1.0 build is tag `v1.0` (1.0-SNAPSHOT) with the same sample compiled in; because a whole-corpus v1.0 run does not terminate, its aggregate was collected per file with a 60 s timeout, and its reported time is a separate warm single-JVM run over the 199 non-hanging files. Open Babel path: `obabel file.cdx -ocan`, dedup on canonical SMILES, then `obabel -ismi -oinchikey`; "unique" = distinct InChIKeys + structures with no InChI (deduped on SMILES). Reproduce with `scripts/perf-overview.sh` and `scripts/perf-overview-obabel.sh`. The corpus includes gitignored `volumeTest/` fixtures present locally; counts vary with the files actually on disk.